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REVIEW

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Wire arc additive manufacturing of aluminium alloys for aerospace and automotive applications: a review

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ABSTRACT

Wire arc additive manufacturing (WAAM) is suitable for printing medium-to-large complex parts with structural integrity while reducing material wastage, and lead time, improving the quality and customized design for functional components. Aluminium alloys are one of the most commonly used metallic materials in manufacturing parts for aerospace and automotive applications due to their lightweight, excellent strength, and corrosion resistance properties. Aluminium alloys have been employed in the WAAM process to produce parts for the aerospace and automotive industries. In this paper, various research works associated with the application of WAAM of aluminium alloys for aerospace and automotive industries, their metallurgical characteristics, and mechanical properties have been reviewed and discussed in detail to identify the research gap and future research directions. This paper is patterned to provide a comprehensive review of WAAM of aluminium alloys for the production of parts in the aerospace and automotive industries.

Abbreviations: AM: Additive manufacturing; Al: Aluminium; Bi: Bismuth; BIW: Body in white; CNC: Computer numerical machines; CMT: Cold metal transfer; CNN: Convolutional neural networks; CL: Curved layer; DE-GMAAM**:** Double-electrode gas metal arc additive manufacturing**;** DWAAM**:** Double wire arc additive manufacturing**;** DMD: Direct metal deposition; DMLS: Direct metal laser sintering; DED-arc: Directed energy deposition arc; 3D: Three-dimensional; EAC: Environmentally assisted cracking; EBM: Electron beam melting; FCI: Fatigue crack initiation; Fe: Iron; GTAW: Gas tungsten arc welding; GMAW: Gas metal arc welding; HE: Hydrogen embrittlement; HAZ: Heat-affected zone; HWAAM: Hot wire arc additive manufacturing; IISCC: Irradiation induced stress corrosion cracking; Li: Lithium; Mg: Magnesium; Mn: Manganese; Ni: Nickel; OL: Online cooling; Pb: Lead; PAW: Plasma arc welding; RS: Robotic system; SCC: Stress corrosion cracking; SLM: Selective laser melting; SCG: Short crack growth; SLC: Super light car; Si: Silicon; Ti: Titanium; Zr: Zirconium

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Introduction

Wire arc additive manufacturing (WAAM) is an essential technique for the fabrication of large-scale aluminium parts using wire as the feed material and an electric arc as the heat source to deposit layer-by-layer 3D components [1]. The use of aluminium alloys in WAAM has become paramount in the aerospace and automotive industries [2–4]. To produce metallic parts, several additive manufacturing (AM)-based technologies have recently been explored by many researchers to manufacture high-performance, customised, and complex metal structures for aerospace, automotive, military hardware, and medical industries [5,6]. Metallic parts can be produced utilising several AM methods (Figure 1) including electron beam melting (EBM), direct metal deposition (DMD), selective laser melting (SLM), direct metal laser sintering (DMLS), and direct energy deposition-arc (DED-arc) which is also known as wire arc additive manufacturing, among others [5–7].

Among the different AM technologies, WAAM technology is being widely accepted by industry due to its potential to fabricate large–Scale, different complex metallic structures. WAAM employs metal welding wire as the feedstock and electric arc as the heat source, which makes it a combination of welding and AM technology. This technology uses welding techniques such as power source, welding torch, wire, and protective gas feeding systems, together with robotic systems (RS) (Figure 2) or computer numerical machines (CNC) which aided the freedom of the welding torch movement for the effectiveness of depositing materials layer-by-layer to form a 3D part [2]. Figure 2 presents a robotic system employed for WAAM [8].

At present, the WAAM technique is being faced with some challenges in the manufacturing process of 3D

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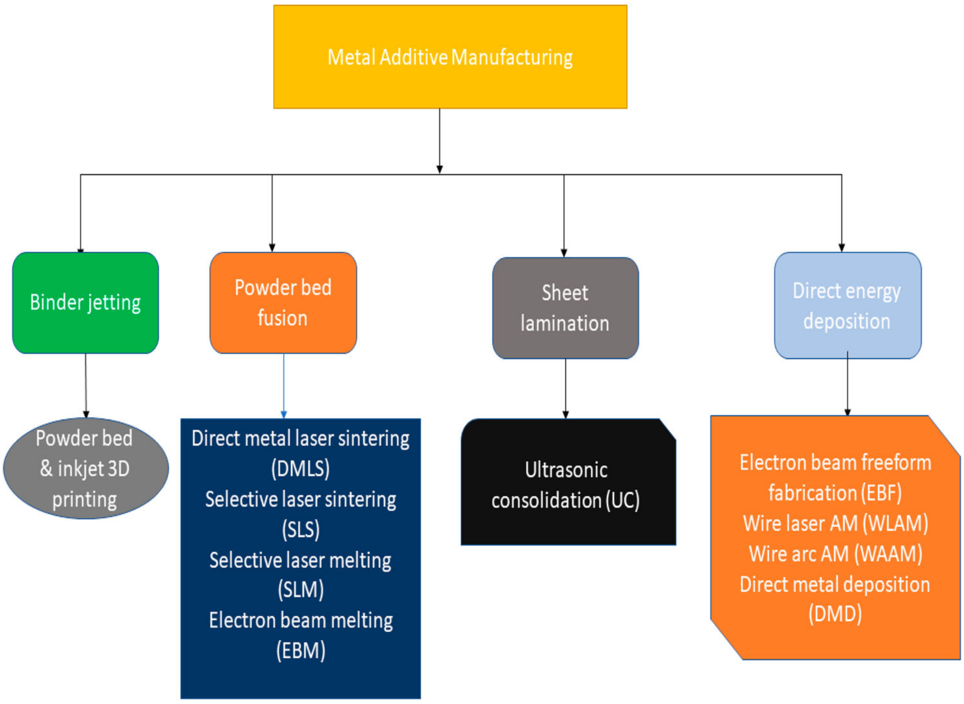

Figure 1. Classification of metal additive manufacturing.

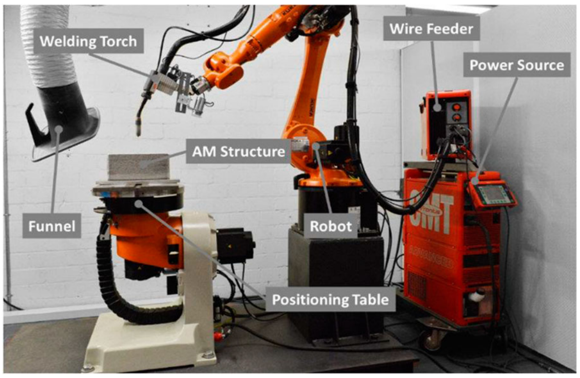

Figure 2. Robotic-guided WAAM system. Reprinted from Ref. [8].

metal parts due to its inherent residual stress and distortion, which are generated by the high thermal input from its heat sources. According to Machirori et al. [9], residual stresses induced by the rapid heating and cooling processes may cause defects such as cracks, distortion, warpage, and delamination. Residual stress and distortion have been widely identified by many researchers as the major challenges facing the WAAM process due to their effects on mechanical properties and component geometric accuracy. One of the influential parameters involved during the preparation of the WAAM procedure requirements is the heat input. According to Rosli et al. [10], the impact of heat input in the WAAM techniques has been described to have significant changes in mechanical properties, grain structure, grain size, and porosity of the parts deposited in the WAAM process. This suggested that the optimum heat input condition during the WAAM process could improve the manufactured parts' mechanical properties, macrostructure, and microstructure. This implies that the reduction of heat input in the WAAM technology can enhance and maintain the geometrical accuracy and deposition rate with an improvement of the microstructure of the fabricated parts. Fahimpour et al. [11] reported that a welding process can be considered successful when the part produced is defect-free with desired microstructure, low residual stresses and distortion levels, and fewer spatter disturbances.

The effect of heat input on the formability, microstructure, and mechanical properties of the WAAM Al-7Si-0.6Mg alloy was investigated by Li et al. [12]. Results demonstrated that the mechanical properties of the WAAM Al–7Si–0.6Mg alloy remain stable over a large range of heat input. It is concluded that the secondary dendrite arm spacing and Fe-phase in the as-deposited alloy gradually increase with an increase in heat input, and slight over-burning occurs in the heat-affected zone at higher heat inputs. After solid-solution and artificially aged (T6 heat treatment) treatments, the size of α-Al grain and eutectic silicon grain increases with the increase of heat input. Hence, it is necessary to investigate the effect of inherent residual stress and distortion generated during WAAM processing as a result of high thermal input from the heat source.

During a layer-by-layer built-up approach during the WAAM process, the heat transferred from the torch to the welding pool and the thermal cycling on each deposited layer has a lot of influence on both the material composition and mechanical properties of the final products [13]. With regard to the application of WAAM aluminium alloys, the choice of a technique with an efficient power source and process parameters has been shown to enhance the microstructure of the fabricated components. Thus, the rationale behind choosing the technique with an efficient power source for a range of materials used in the WAAM process has to be extensively studied. According to the different power sources, the WAAM technology can be divided into gas tungsten arc welding (GTAW), gas metal arc welding (GMAW), and plasma arc welding (PAW) [10]. Owing to increasing demand for flexibility and product integration, combined with reduced product development cycle and robust mechanical properties, WAAM of aluminium alloys for aerospace and automotive applications has been well-documented in terms of processability, metallurgical characteristics, mechanical properties, and resulting material properties.

WAAM method

Many researchers are successfully utilising arc-based welding technologies in WAAM. The welding techniques employed in WAAM are GMAW, GTAW, and PAW. The selection of the welding method for WAAM depends on the type of material and applications. In GMAW, the source of heat is an arc formed between a consumable metal electrode and the workpiece that melts together to create a weld pool that fuses to form a joint [14]. The rate at which the material transfer from the tip of the consumable electrode into the weld pool has a significant impact on the overall performance of GMAW as it influences process stability, spatter generation, weld quality, and the positional capabilities of the process (Figure 3).

In arc-based processes like GMAW, deposition rates of up to 8 kg h^{-1} can be actualised [16,17]. When compared with GTAW- and PAW-based WAAM, GMAW-based WAAM has approximately 2–3 times high deposition rate [18]. There are four primary methods of metal transfer in GMAW in which each has distinct features. These include globular, short-circuiting, spray, and pulsed-spray metal transfer methods [19].

One of the significant developments in the WAAM system came into the limelight when the cold metal transfer (CMT) process has been integrated into the place of GMAW. The CMT process is an improvement of the GMAW process with less heat input, less spatter disturbances, and better weld quality with a high deposition rate as discussed by Boțilă [20]. It is usually performed by a welding robot, applying a mechanism that has a controlled plunge transfer mode. Specifically, the welding of aluminium and steel with better results has been excellent with the use of the CMT welding process [21]. Wu et al. [22] examined the effect of the cooling arc CMT process on AM. The results showed that CMT mode was the most stable process, and CMT plus pulse (CMT + P) mode with an alternating of CMT and pulse processes had a finer grain and bigger hardness. Feng et al. [23] found out that the CMT process is suitable for welding thin aluminium alloy sheets due to the low heat input and the slight deformation. In another investigation, Prado-Cerqueira et al. [24] reported the problem that could be discovered from the use of GMAW equipment as thermal input. This excessive thermal input can lead to deformation in both the substrate and the weld beads. To overcome this challenge, the CMT welding technique has been proposed by many researchers as an effective method to subdue this problem [24]. In addition, the effect of porosity and cracks in WAAM of fabricated parts can be reduced or eliminated by adopting advanced techniques such as CMT. The presence of pores in the built-up samples in the WAAM process has been reported by Veiga et al. [25], as the

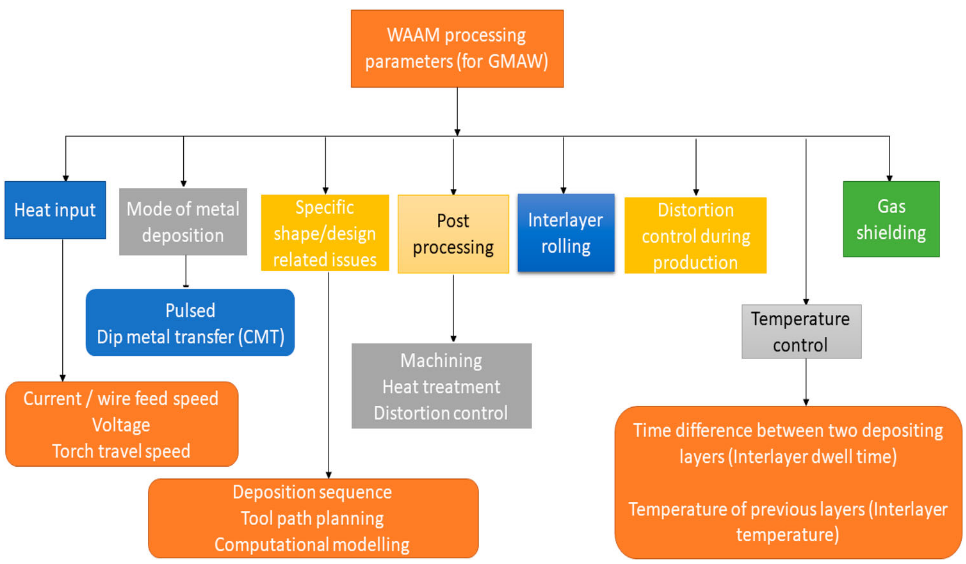

Figure 3. Parameters affecting WAAM product quality. Reprinted from Ref. [15].

main challenge facing GMAW deposition. The study further demonstrated that the level of pores generated through GMAW deposition can be reduced with alternating negative and positive intensity energy cycles or the alternating current (AC) mode.

In GTAW, filler material, shielding gas, and heat source have been widely regarded as the significant basic requirements needed for the GTAW process [26]. GTAW uses a non-consumable electrode to produce the arc and the filler material is fed externally with a shielding gas applied to protect the arc and the weld pool area [27,28], which is different from GMAW, as it employs the filler material as the electrode. GTAW uses tungsten as the electrode to produce arc [26].

GTAW is complex, and it is significantly slower than most other welding techniques. To improve the workability and controllability in GTAW, an innovatively modified filler metal and arcing-wire GTAW process were experimentally demonstrated by Chen et al. [29]. It was reported that the experimental data and the comparative analysis suggested that the arcing process and the deposition speed improve the measure of the weld controllability. Hence, further research studies are expected to cover the advanced techniques that can be used to improve the workability and controllability in GTAW to outperform the conventional GTAW and GMAW processes.

PAW is a process similar to GTAW as the arc is formed between a pointed tungsten electrode and the workpiece [30]. At present, major limitations of a PAW process include noisy operation, harmful irradiation, and higher cost of PAW maintenance. In the PAW process, flow rates for shielding gases are usually in the range of 5–15 L min^{-1} for low–current applications. For high–current welding, flow rates of 15–32 L min^{-1} are used. PAW provides high-quality welds with very little distortion at a high torch travel speed and high energy density arc, but it requires high capital expenditure as compared to GTAW and GMAW [20,31], and the ability to produce lightweight components from costly materials such as aluminium and titanium without material waste. This makes arc and plasma AM interesting for many commercial industries, in particular aerospace, automotive, and medical industries.

Among the different WAAM processes, GMAW in the CMT process mode has been regarded as a suitable choice for aluminium alloys. The CMT utilises short circuit welding to detach single droplets of feedstock wire into the weld pool. Whereas GTAW and PAW utilise a non-consumable tungsten electrode to produce the weld. This is different from the CMT, the wire is fed through the arc with a variable direction oriented to the substrate surface. Moreover, the feeding of filler wire through variable orientation direction during GTAW and PAW processes affects the characteristics and quality of the deposited product as reported by Ding et al. [19].

Recent aluminium alloys manufactured using WAAM

For enhancing energy efficiency and mechanical performance, new novel aluminium alloys are now being developed and manufactured using WAAM techniques for the applications of aerospace and automotive industries. These new aluminium alloys have been developed by alloying them with new elements, such Silicon (Si), Manganese (Mn), Magnesium (Mg), Zinc (Zn), Iron (Fe), Chromium (Cr), Nickel (Ni), Titanium (Ti), Zirconium (Zr), Lithium (Li), Lead (Pb), Bismuth (Bi), Copper (Cu), etc., to exert robust mechanical properties and also to improve the printability of the printed parts.

Gu et al. [32] applied inter-layer rolling to the WAAM Al–Mg4.5Mn alloy with increasing loads of 15, 30, and 45 kN to achieve excellent mechanical properties for 5087 (Al–Mg4.5–Mn) alloys. It was reported that the combination process of rolling deformation with WAAM deposition is referred to as an effective method in refining microstructure and enhancing mechanical properties for AM aluminium alloys.

Radovic et al. [33] observed deformation behaviour and microstructure evolution of AlMg6Mn alloy during shear spinning. The result showed that the addition of alloying elements contributed to grain refinement and dislocation reactions with particles and atoms of Mg and Mn in solid solution.

In another study, Panchenko et al. [34] developed a high-performance-controlled short-circuiting metal transfer process at a wire feed rate of 12 m min^{-1} for WAAM with an Al–Mg–Mn alloying system. The result showed an improvement in the mechanical properties of the fabricated parts with 41% elongation of the tensile samples.

Gu et al. [35] examined the effect of inter-layer cold working and post-deposition heat treatment on the additively manufactured Al–6.3Cu alloy. The ultimate tensile strength and yield strength of the inter-layer rolled alloy with 45-kN load are 314 and 244 MPa, respectively. After T6 treatment, the UTS and YS of both of the as-deposited and 45-kN rolled alloys surpassed 450 and 305 MPa, respectively, which are higher than the properties of the wrought 2219 T6 alloy.

Qi et al. [36] examined the double-wire plasma system for WAAM of Al–6.3Cu alloy and added Mg into Al–Cu alloy to enhance the mechanical properties. The amount of Cu and Mg was optimised with the wire feed speed. The ultimate tensile strength, yield strength, and elongation of the deposit in horizontal direction were 176MPa, 76.6MPa, and 11.4%, respectively. Moreover, the mechanical properties of the sample's built-in vertical direction were slightly lower than the sample's built-in horizontal direction.

An et al. [37] researched the effects of vanadium content on the microstructure, tensile properties, and impact toughness of Ti–6Al–xV components produced by WAAM. It was reported that a monotonic increase in yield strength and ultimate tensile strength was observed with the increase of vanadium content while the fracture strain and impact toughness changed in the opposite trend. In addition, irregular-shaped, plate-like features without columnar grains and heat-affected zone (HAZ) banding were obtained in Ti–6Al–xV alloy.

In another investigation, Vishnukumar et al. [38] utilised WAAM for repairing aluminium structures in marine environments. Being a marine-suitable Al grade when exposed to sea water regularly, its corrosion-resistant properties decline over time. To restore the corroded surfaces, it was reported that aluminium filler material ER4043 was deposited on AA5052 using CMT process-based WAAM technology.

3D-printing of aluminium components by WAAM

Aluminium alloys are widely used in different industrial applications such as aerospace, shipbuilding, and automotive industries because of their lightweight, excellent mechanical properties, and high corrosion resistance [5,6]. WAAM is capable of manufacturing high-strength components with complex design and geometric accuracy as it is done by layer-by-layer addition of material. WAAM often displays outstanding benefits in the manufacturing of complex parts.

As a result of this, there is keen interest from researchers to manufacture aluminium-based structures using WAAM by controlling the various deposition parameters for the production of aerospace and automotive spare parts. The usage of various aluminium alloys in WAAM is currently gaining awareness among many researchers [39–41]. Recently conducted research shows that the most promising alloys are Al–Li [42], Al–Cu [43,44], Al–Mg [45–47], Al–Zn–Mg–Cu [48], Al–Cu–Mg [42,43], Al–Si–Mg [45], Al–Mg–Si [44], and Al–Mg–Mn [46,47]. Among all these categories of various aluminium alloys, Al–Mg–Mn and Al–Zn–Mg alloys have shown significant improvement due to their high strength and corrosion resistance properties. Moreover, the application of these alloys is widely used in shipbuilding, aerospace, and automotive industries as the main manufacturing materials [47,48], which makes it reasonable to use Al–Mg–Mn and Al–Zn–Mg alloys in WAAM of high-value parts.

To further improve the mechanical performance of WAAM Al alloys parts, there is a need to analyse the influence of environmentally assisted cracking (EAC) on the mechanical properties of the corrosion-resistant metallic parts fabricated by the WAAM method. EAC often influences a wide range of failure in materials such as stress corrosion cracking (SCC), hydrogen embrittlement (HE), sulphide stress cracking (SSC), and irradiation-induced stress corrosion cracking (IISCC). Experience has shown that aircraft components made from Al 7075-T6 tend to corrode rapidly, particularly in aircraft operating in marine environments [49]. The wear and corrosion behaviour of the WAAM-fabricated aluminium alloys need to be properly examined because the aluminium alloy-based components are highly utilised in the area where it encounters wear and corrosion during its service life [47]. Excessively corroded parts have to be replaced for visible safety reasons and the replacement costs are high. Hence, there is a need to fully investigate EAC behaviour in WAAM Al alloy-fabricated parts.

WAAM technology has witnessed appreciable industrial growth recently, while basic properties such as impact toughness, tensile strength, and hardness are widely reported. Despite the robust advantage that came with the technology of WAAM techniques, there are still some potential challenges associated with the processing of Al alloys using the WAAM system. However, aluminium can be further alloyed with other metals to subdue some of these challenges [50]. Langelandsvik et al. [51] investigated WAAM with TiC-nanoparticle reinforced AA5183 Al–Mg alloy. By the incorporation of nanoparticles in the feedstock wire, TiC-modified AA5183 Al–Mg alloy exhibited fine-grain refinement with the attainment of improved strength. The hardness increased by 13% with the addition of TiC. In another study, Hauser et al. [52] examined pore formation in WAAM of AW 4043/AlSi5(wt-%). It was reported that the large number of pores exhibited in aluminium parts is in relation to the shielding gas flow rate, since at a higher gas flow rate, the melt pool solidifies faster as a result of increased convective cooling, thus preventing the process-related gas inclusions from escaping. Since a lower gas flow rate leads to reduced convective cooling, the melt pool remains as a liquid for a longer period allowing trapped gas in the melt pool to escape, thus reducing porosity. Zhou et al. [53] mixed Nb powder with a mean particle size of 15 μm to the molten pool during WAAM of 5B06 aluminium part. It was observed that fine equiaxed grains with a mean size of lesser than 50 μm were attained. An improvement in both the ultimate tensile strength and elongation of the Nb-added samples with enhanced micro-hardness was reported due to the refined grain structure.

The parts manufactured by WAAM require performance characteristics with the quality of the deposit, surface finish (Table 3), geometry accuracy, including the microstructural features (grain structure, grain size, texture, etc.), and resultant mechanical properties (hardness, tensile strength, impact strength, yield strength, compression strength, etc.).

Enhancement of the mechanical properties of WAAM aluminium alloys is highly demanding in recent times. The mechanical properties and quality performance of the AM aluminium parts fabricated by the WAAM system need more improvements for structural applications. The production of lightweight structures using WAAM techniques to attain robust mechanical properties will not only enhance aircraft performance but also increase its acceleration efficiency. Several research works have been extensively carried out to improve the mechanical properties and structural performance of the WAAM-fabricated aluminium parts by varying different manufacturing parameters, which include arc mode [54], wire feeding speed [51], speed of the torch [51,54–58], and by applying post-process treatment [59] (Table 1).

Therefore, lots of research work related to WAAM of aluminium alloys has been published to characterise various types of mechanical properties (Table 4) and microstructural performance (Table 2) of WAAM-processed Aluminium alloys under different manufacturing conditions. Table 4 consists of the various mechanical properties examined by researchers for different types of WAAM-processed aluminium alloys, namely Al–Mg, Al–Zn–Mg, Al–Mg–Mn, Al–Cu, Al–Mg–Si, Al–Si, Al–Cu–Mg, Al–Mg–Sc, Al3Nb, Al 7055, Al 4043, Al 5356 and Al 5183.

To produce highly valued parts using WAAM technology, aluminium can be further alloyed with other alloying elements to improve its processabilities and mechanical performance [50]. As for the tensile strength, Table 4 reveals that some Al alloys like Al–Zn–Mg [55], Al3Nb [53], Al–Cu [35], Al–Mg–Mn [32], Al–Mg–Sc [67], Al–Mg–Si [12], Al 5183 [70] exhibited a high tensile strength of about 324, 360, 450, 344, 363, 354.5, and 293 MPa, respectively. As presented in Table 4, the yield strength of the selected WAAM Al alloys such as Al–Zn–Mg [55], Al–Mg–Mn [32], Al–Mg–Sc [67], Al–Cu [35], and Al–Mg–Si [12] indicated a high yield strength of 208, 240, 258, 305, 310 MPa respectively, while Al3Nb [53], Al 5183 [70], Al–Mg–Sc [67], Al–Mg [62], and Al 5356 [25] showed appreciable elongation of 20%, 20%, 26%, 27%, and 33%, respectively.

It is well-known that manufactured parts used in aerospace and automotive applications are frequently exposed to operating conditions such as high fatigue, impact, wear, high-stress conditions, and high variable temperature through their daily usage in service. Hence, it is necessary to determine the mechanical response of the different combinations of alloying elements in aluminium parts produced by WAAM under different loading and operating conditions.

Table 1 highlights a key published research literature conducted on WAAM of aluminium alloys, which are reviewed in detail in this study, presenting the main findings of each article, considering the process conditions (such as welding speed, current, voltage, build orientation, and deposition strategy), types of mechanical properties, and AM machine used. It was also noted that most commercial industries have been exploring WAAM machines to additively manufactured aluminium alloyed parts for the benefit of the aerospace and automotive industries. Tables 2–4 present the data of microstructural, surface quality, and mechanical properties of aluminium alloy fabricated from the WAAM process.

Current applications of aluminium alloys produced by WAAM in aerospace and automotive industries.

Current breakthrough in WAAM technology has made it possible to attract significant industrial attention in recent times due to its ability to produce functional parts which include impeller shaft and aircraft landing gear ribs (Figure 4) with a weight saving of approximately 78% in raw material when compared with the conventional subtractive manufacturing process [28]. The aerospace and automotive industries require manufactured parts with enhanced mechanical properties which meet operating standards set for their usage.

The possibility of employing WAAM technology for the mass production of spare parts for the replacement of damaged components in the aerospace and automotive industries is highly laudable, as this will reduce delivery lead time and scarcity of spare parts in the workshop for the maintenance of aerospace and automotive equipment and facilities for improved efficiency and performance. A lot of parts, as well as spare parts, can be manufactured by using WAAM when necessary, thus eliminating the need for storage or remote transport of them. The quality of the parts built with the WAAM-manufactured aluminium alloys for the production of spare parts in aerospace and automotive equipment and facilities can be further improved by incorporating the various sensors with the WAAM system to measure welding signals [82], metal transfer behaviour [83], deposited bead geometry [84], and interpass temperature [85], thereby supporting in-process monitoring and control to attain higher product performance [17].

According to Zhu et al. [86], the relatively high specific strength and stiffness, good ductility and corrosion resistance, low price, and excellent manufacturability and reliability make advanced aluminium alloys a popular choice of lightweight materials in many aerospace and automotive structural applications. The application of lightweight structures brings benefits to aircraft performance, e.g. increased energy efficiency, acceleration performance, payload, flight endurance, and reduced life cycle cost and greenhouse gas emissions. The advantages of using WAAM for the production of spare parts in the aerospace and automotive industries can be briefly described as follows:

(a) The possibility to create 3D components with large complex parts, which can be difficult or impossible

Table 1. Summary of the process conditions and mechanical properties of WAAM-manufactured aluminium alloys.

Material	Process conditions	Mechanical properties evaluated	Main findings	Machine	Ref.
Al–Zn–Mg	(a) Welding speed = 20–35 cm min^{-1} (b) Current = 97–112 A (c) Voltage = 16.1–17.4 V	Tensile strength, yield strength, elongation, and hardness	(a) Increased tensile strength, yield strength, and elongation for the horizontal samples (b) Enhanced microhardness value for the as-deposited material (c) The WAAM-manufactured parts exhibited a mixed ductile–brittle fracture	WAAM	[55]
Al–Mg–Mn–TiC	(a) Current = 85 A (b) Voltage = 11.9 V (c) Travel speed = 0.48 m = min (d) Wire feed speed = 5.8 (m = min)	Tensile strength, hardness	(a) TiC-modified AA5183 Al–Mg alloy exhibited grain refinement compared to a commercial AA5183 benchmark (b) Increased hardness by 13% with the addition of TiC	WAAM	[51]
Al–Mg–Mn	(a) CMT = 320 min, (b) CMT-ADV = 250 min, (c) CMT-PADV = 215 min	Tensile strength, hardness	(a) Enhanced hardness for the CMT-PADV (b) Lower tensile strength in the vertical direction	WAAM	[60]
Al–Mg–Mn	(a) Polarity variation = 10:10, (b) Mean wire feed rate = 4.5–5.2 m min^{-1} (c) Deposition rate = 08.1-0.94 kg h^{-1} (d) Current I/Imax (A) = 76/109:86/107:74/94 A, (e) Mean voltage U = 8.6:12.4:10.6 V, (f) Travel speed = 12, (g) Energy input per unit length = 56:90:67 (J mm^{-1}).	Macro- and microstructure, and mechanical properties	(a) Increased surface quality of the final specimen due to the control of the heat input during the intrinsic heat treatment	WAAM	[61]
Al–Mg	(a) Deposition strategy (circling and hatching), (b) Shielding gas, (c) Gas flow rate = (30 L min^{-1}, 25 L min^{-1}, 18 L min^{-1})	Tensile strength, yield strength	(a) A resultant porosity below 0.035% in both Ar + circling and Stargold + hatching cases (b) High porosity levels (up to 2.86 area %) obtained in the worst conditions, which had a reduced impact on the static tensile test	WAAM	[62]
Al–Zn–Mg	(a) Arc mode = (CMT, CMT + P, CMT + A, CMT + PA), (b) Wire feeding speed = 0.6 m min^{-1}, (c) Deposition speed = 6 m min^{-1}, (d) Shielding gas flow = 20 L min^{-1}	The tensile strength and yield strength	(a) The tensile and yield strength of the CMT + PA sample show the optimum tensile properties among the samples fabricated by different arc modes	WAAM	[54]
Al–Mg	(a) Current = 30–50A, (b) Arc voltage = 14–15 V), (c) Welding speed = 12–15 mm s^{-1}, (d) Wire feed rate = 2–3 m m^{-1}, (e) In Deposition rate = 0.5–2 kg h^{-1}) (f) Layer height = 0.5–2 mm, Electrode to layer angle = 90° (g) Wire diameter = 1.2 mm	Tensile strength, hardness	(a) Highest tensile properties obtained in the L direction (b) Lowest mechanical properties obtained for the T direction	WAAM	[56]
Al–Cu and Al–Mg	(a) Inter-layer rolled = 5 KN, 10 KN, 45 KN. (b) Rolled heat treated = 45 KN	Tensile strength and microstructure properties	(a) Micropores number, volume, size, and roundness in rolled alloys decreases with increasing loads	WAAM	[59]
Al–Mg and Al–Si	(a) Open circuit voltage = (15.2V, 15.3V, 18.0V) (b) Arc current = (156A, 177A, 115A) (c) Travel speed = (8 mm s^{-1}, 6 mm s^{-1}, 6 mm s^{-1}) (d) Process efficiency = (0.9, 0.9)	Physical properties (thermal expansion coefficient, thermal conductivity, specific heat capacity, temperature dependent elastic modulus	(a) WAAM depends on setting heat source modification for every layer is an effective way to conduct WAAM properly (b) Improve deposition efficiency and manufacturability via adequate heat source settings	WAAM	[63]
Al–Si	(a) Uni-directional (b) Bi-directional strategy	Tensile strength, hardness	(a) Optimum WAAM sample was achieved via bi-directional strategy and operating with lower heat input process parameters	WAAM	[64]

(*continued*)

Table 1. Continued.

Material	Process conditions	Mechanical properties evaluated	Main findings	Machine	Ref.
Al–Cu–Mn	(a) The bypass current ratios with = 0%, 50%, 70%, 90%. (b) Bypass current = 0A, 80A, 112A, 144A (c) Arc voltage = 22.5V (d) Travel speed = 0.6 m min^{-1} (e) Total current = 160V.	Ultimate tensile strengths, microhardness	With bypass current ratio from 0% to 70% (a) Increase in ultimate tensile strengths was obtained (b) Enhanced elongation.	Double-electrode gas metal arc additive manufacturing (GE-GMA)	[65]
Al–Zn–Mg–Cu	(a) EP Current = 220 A, (b) EN Current = 120 A, (c) EN-EP Balance = 80 % (d) AC Frequency = 105 Hz, (e) Arc Length = 4 mm	Microhardness, tensile strength, yield strength	(a) The average yield strength and ultimate strength are higher in the vertical direction than in the horizontal while the difference in elongation is small	WAAM	[57]
ER-4043 Aluminium	(a) Gas Pressure = 12, 10, 14 bar (b) Current = 55, 60, 65 Ampere (c) Nozzle distance = 6, 8, 10 mm	Tensile strength, hardness,	(a) The A3, B3 & C2 specimen has maximum tensile strength while A3, B3 & C3 specimen has maximum Hardness	WAAM	[66]
Al–Mg–Sc	(a) Current (I) = (49A, 102 A) (b) Arc voltage (U) = (14.5V, 18.3 V) (c) Travel speed (TS) = 10 mm s^{-1} (d) Wire feed speed = (3 mm min^{-1}, 6 mm min^{-1})	Tensile strength, yield strength	(a) High tensile strength, yield strength, and elongation of the horizontally oriented specimens	Double-Wire Arc Additive Manufacturing	[67]
Al–Si	(a) Travel speeds = 600 mm min^{-1}, 1200 mm min^{-1}, 2400 mm min^{-1} (b) Average power = 1039 W, 1470 W, 1660 W,	microstructure and morphological aspect of the deposited weld beads.	(a) CMT parameters leads to the understanding of the metal transfer process, which could be used to control minimal variations of a fixed configuration in order to improve the geometrical characteristics of the multilayer deposit	WAAM	[68]
Al–Cu	(a) Diameter = 3 mm (b) Wire diameter = 1.2 mm, (c) Wire feed speed = 1.5 m min^{-1} (d) Shielding gas: argon = 99.99% purity, (e) Shielding gas flow rate = 15 L min^{-1} (f) Travel speed = 0.3 m min^{-1}	Microstructure, residual stress, and tensile properties	(a) Decrease in grain size (b) Increased percentage of grains with low angle boundaries (c) Modified residual stresses from tensile to compressive state	WAAM	[69]

to achieve through conventional manufacturing processes [87,88].

(b) The possibility of use in a wide range of materials [87–89].
(c) Flexibility in making complex parts [90].
(d) The possibility to create high-quality parts, while maintaining the mechanical strength and at the same time benefiting from a lower weight of the components [87,91].
(e) The increasing demands of aluminium parts, particularly alloy with high strength in the aerospace and automobile domain, can be actualised with the use of WAAM as a future economic opportunity [92]. Figures 4 and 5 present some aluminium products fabricated with the WAAM process.

Use of WAAM on aluminium alloys in aerospace and automotive industries

In recent years, different research groups are concentrating their studies on the usage of aluminium alloys for the WAAM system [44,60]. Different range of aluminium alloys are being examined by many researchers using the WAAM process to enhance the production of spare parts in the aerospace and automotive industries. High-strength Al–Zn–Mg–Cu alloys are widely utilised in the aerospace industry, owing to their high strength-to-density ratio and competitive cost. WAAM has the potential to create aluminium components with a robust design and improved mechanical properties [32,93–95]. The usage of aluminium alloys in WAAM has attracted huge attention in the aerospace and automotive industries, especially for the production of structural parts of airplanes [39,41]. Aluminium alloys are of particular interest due to their high strength (Figure 6), low mass density, excellent ductility, and high corrosion resistance; high electrical and thermal conductive properties and very good manufacturability (such as extrusion, rolling, bending, welding, repairability, and AM) render aluminium alloys the most promising aerospace and automotive materials [5,6]. Josten and Höfemann [97] utilised arc welding-based AM for body reinforcement in automotive engineering. It was reported that the WAAM process can be used to reinforce body components in automotive engineering by generating stiffening elements. In automotive industries, WAAM could be used for creating local stiffening elements to reinforce critical body parts

Table 2. Microstructure properties aluminium alloy WAAM-manufactured component.

Material	Process	Condition	Microstructure	Machine	Ref.
Al 5087	CMT	AF	Coarse grain structure	WAAM	[32]
		Rolled (45KN)	Fine-grain structure rolling texture		
Al 5356	GMAW	Ar shield	Al and Al_3Mg_2 phases	WAAM	[71]
		N_2 shield	Al, Al_3Mg_2 and AlN phases		
Al 5183	CMT	AF	Smaller columnar grain	WAAM	[72]
	CMT + P	AF	Smaller columnar grain		
	CMT + ADV	AF	Coarser columnar grain		
Al 4220	CMT	T_6 heat treatment	The size of the α-Al and eutectic silicon grains increased	WAAM	[73]
Al 5356	CMT + P	AF	Evenly disperse fine porosity (less than 50 µm)	WAAM	[8]
Al 5356	CMT	AF	α-phase (Al) and β-phase (Al_3Mg_2)	WAAM	[74]
Al 5556	CMT,	AF	The average grain size measured in the CMT process was 63.3 µm in the lower wall section	WAAM	[60]
	(CMT-ADV)	AF	The multilayer samples welded using the CMT-ADV and CMT-PADV arc modes with alternating polarity showed a significantly finer microstructure with a cellular grain structure compared to the welded sample using the standard CMT method		
	(CMT + PADV)	AF			
Al 5056	CSC	AF	Smaller Coarse grains. Columnar grains are located in the lower part of each deposited layer and fine grains in the upper part	WAAM	[34]
Al 2024	CMT + PADV	AF	The microstructure of the as-deposited alloy was characterised by a hierarchical distribution of dendrites, equiaxed grains, and a slight number of columnar grains	WAAM	[43]

Key: AF: As-fabricated; CMT: Cold metal transfer; P: Pulse; ADV: Advanced; Controlled short-circuiting (CSC) metal transfer.

Table 3. Surface quality of aluminium alloy WAAM-manufactured component.

Material	Condition	Main findings	Machine	Ref
Al–Si	(a) Wire filling speed = 5.5 m min$-^1$ (b) Scanning speed = 8 mms$-^1$ (c) Arc current = 127 A (d) Arc voltage = 17.5 V (e) Feed rate = 25 mm s$-^1$ (f) Spindle speed = 12,000 r min$-^1$ (g) Milling thickness = (0 mm, 0.4 mm, 0.8 mm, 1.2 mm, 1.6 mm).	When the milling thickness (t) is at the range of 0.4–1.2 mm. The surface roughness and manufacturing allowance of the wall was reduced by 22.9% and 31.6% respectively, as compared to the component deposited by WAAM	WAAM/ Milling system	[75]
Al–Mg	(a) Wire feed speed (WFS) = 4.9 m min^{-1} (b) Shielding and trailing gas = 4.5 (c) Ar, shielding gas flow (GFR) = 15 L min^{-1} (d) Contact tip-to-workpiece distance (CTWD) = 15 mm	It was stated that an increase in torch speed over the higher limit can result in top surface undulation and a decrease in torch speed below the lower limit can lead to surface waviness. However, the surface waviness of the component can be reduced by pre-heating the substrate to a suitable temperature	WAAM	[76]
Ni-based superalloys	(a) Travel speed = 78 to 90 mm min^{-1} (b) Deposition current = 22 to 26 A (c) Feed rate = 4 to 5.5 g min^{-1}	It was demonstrated that when the substrate is heated to 300 °C, the surface waviness was significantly decreased due to the increased wettability of each layer	WAAM	[77]

[97]. Thapliyal [47] reported that the product developed using WAAM exhibited robust mechanical properties which are better than the wrought or cast products of aluminium. Hirsch [96] studied recent development in aluminium for automotive applications. It is concluded that aluminium is the ideal lightweight material for the production of car body parts (such as body-in-white (BIW), hoods, doors, wings, bumpers, and interiors) as it allows a weight saving of up to 50% without compromising safety (Figure 7) [98]. In another study, Botila [20] considered that the use of aluminium alloys (such as Al–Cu–Mg, Al–Li, Al–Cu-Sn, Al–Si–Mg, and Al–Mg) in the construction of aircraft structural elements reduces their total mass and thus allows them to become more efficient in terms of fuel consumption (as shown in Figure 8). Zhang and Liang [99] studied metal AM in aircraft, current application, opportunities, and its challenges. It is concluded that metal AM provides a new solution in terms of weight, cost, and time reductions for the production of spare parts in the aircraft manufacturing industry.

Challenges facing the application of WAAM in aerospace and automotive industries

The application of WAAM aluminium alloys has been constrained by common defects which include porosity defect [57,63,77], oxidation (surface oxidation and oxidation anomalies) [59], humping effect [63,77], cracking [101,102], distortion [103], high levels of residual stress [77,103], geometrical accuracy, delamination, and deformation [104]. These defects usually affect the

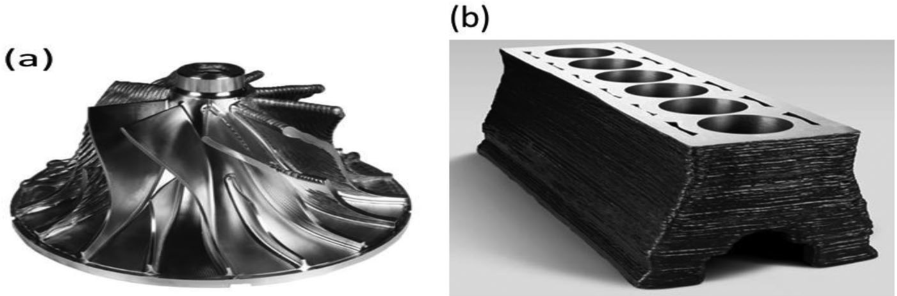

Figure 4. (a) Impeller shaft produced by Harlow-FasTech using a combination of WAAM and machining. Reprinted from Ref. [93], (b) An example of a part printed by Lincoln Electric via WAAM. Reprinted from Ref. [94].

Table 4. Mechanical properties of aluminium alloys fabricated from WAAM processes.

Aluminium alloys	Condition	Tensile strength (MPa)	Yield strength (MPa)	Elongation %	Ref.
Al–Zn–Mg	HOS	324	208	9.66	[55]
Al–Mg	AF	270	110	27	[62]
Al–Zn–Mg	VOS	299	188	8.98	[55]
Al–Cu	AF	260	132	15.5	[78]
Al_3Nb	AF	360	-	20	[53]
Al–Mg–Si	HOS	176	76.6	11.4	[36]
Al 5183	AF	293	145	-	[70]
Al–Cu	AF	314	244	-	[35]
Al–Cu	AHT	450	305	-	[35]
Al–Mg–Mn	ILR	344	240	-	[32]
Al 4043	PF	108.6	40	14.5	[79]
Al 7055	AF	230	148.3	3.3	[57]
Al 5356	VOS	264	106	27	[25]
Al 5356	HOS	277	111	33	[25]
Al–Mg–Sc	HOS	363	258	26	[67]
Al 5356	HOS	267	114.9	28.3	[8]
Al 5356	VOS	253	114.9	17.1	[8]
Al–Cu	VOS	258	106	15.5	[80]
Al–Cu	HOS	263	114	18.3	[80]
Al 5183	AF	293	145	20	[70]
Al 4043	AF	151.91	69.7	16.8	[81]
Al–Cu–Mg	OH	293	185	12	[43]
Al–Cu–Mg	AHT	245	185	4	[43]
Al 5183	VOS	293	145	-	[70]
Al–Mg–Si	AF	354.5	310	6.3	[12]

VOS: Vertically oriented built sample; AF: As-fabricated; HOS: Horizontally-oriented built sample; AHT: After heat treated; ILR: Interlayer rolling; PF: Pulse frequency.

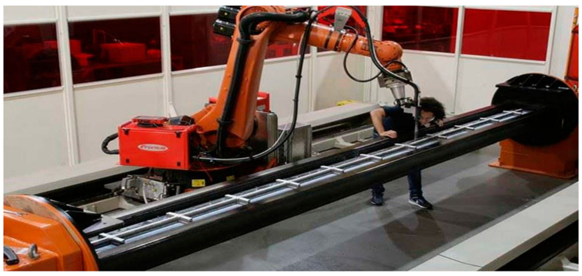

Figure 5. Spar made of aluminium alloy for the aerospace industry, using robotised WAAM. Reprinted from Ref. [95].

mechanical performance of aluminium alloys WAAM-manufactured parts in service.

Residual stress and distortions or deformation

The residual stress results in significant plastic deformation, leading to warping and distortion, loss of geometric tolerance, delamination of layers during deposition, and deterioration of fatigue performance of WAAM-processed products [104]. According to Jarari et al. [77], the residual stress has been regarded as a major cause of distortions, the loss of geometrical accuracy in WAAM components, due to large thermal gradients during repeated melting and cooling processes, besides its influence on mechanical properties and grain structures [105]. The distortions are caused by thermal expansion and shrinkage of the part during repeated melting and cooling processes, which is a major challenge for large thin-walled structures [88].

Porosity

One of the challenges facing the acceptability of WAAM for aluminium component production is high levels of porosity [105]. As described in Ryan et al. [106], many factors might affect porosity, including wire feed speed, travel speed, metal transfer mode, and wire batch. WFS refers to the rate at which the wire is fed to the arc and is influenced by current and voltage. TS refers to the rate that the arc moves to the workpiece.

In another study, Horgar et al. [70] pointed out other factors that might influence the porosity in aluminium welds such as cleaning of support plate and wire, wire surface quality, shield gas cleanliness, welding parameters, and polarity. Gu et al. [107] described that the surface finish of the wire could significantly affect porosity as wires with poor surface finish result in unstable arcs during WAAM, and unstable arcs are related to porosity.

About WAAM of aluminium alloys, Horgar et al. [70] and Cong et al. [108] reported that one of the major issues for aluminium alloy is pore formation in the built parts.

Humping defect

Humping is a common defect in WAAM samples fabricated using GMAW [109]. Humping defect also plays a major role in the reduction of hardness and tensile strength of the fabricated WAAM sample [110,111]. As

Figure 6. Aluminium high strength aerospace components. Reprinted from Ref. [100].

discussed by Wu et al. [112], three factors are responsible for the humping formation. The first factor is the high momentum of the backward fluid flow, which causes the initiation and growth of swelling. The second factor is the large variation of the capillary pressure of the liquid channel in the welding direction, which can promote the shrinkage of a liquid channel. The final factor is the capillary instability, making the weld pool unstable and susceptible to collapse [112].

Koli et al. [110] described that in the GMAW process, the humping defect happens due to inappropriate selection of process parameters, heat input, travel speed, surface condition, and chemical composition of the base. It is concluded that the humping defect can be controlled by using low heat input parameters which ultimately improve the mechanical properties of WAAM-manufactured components [110].

Cracking

Cracking is another common defect that is often observed in aluminium alloy fusion welds and is a function of how the metal alloy solidifies. The crack is located in the high-temperature part of the reheated area with equiaxed grains and occurred in some other beads as well [70]. According to Koli et al. [110], tensile residual stress causes crack nucleation and further

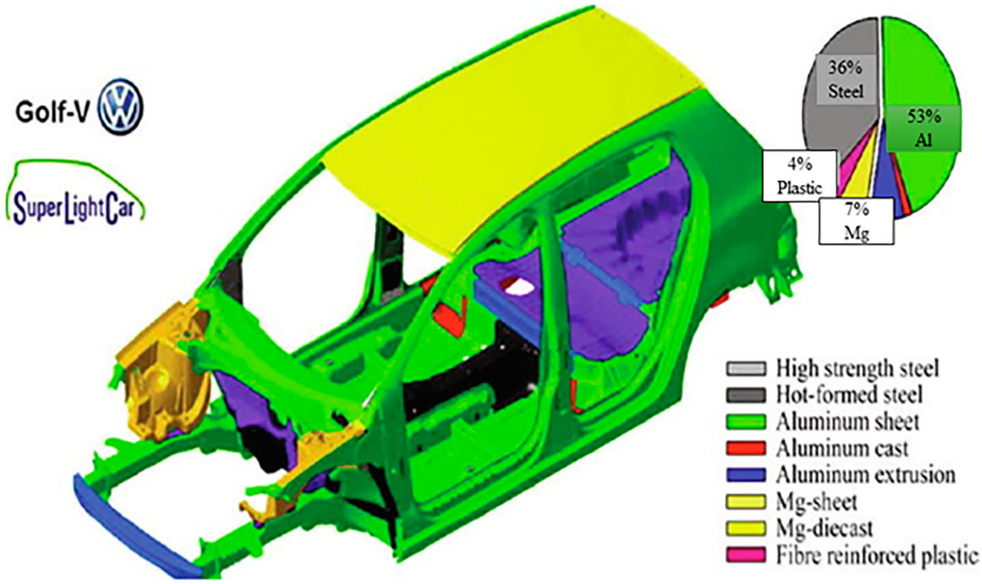

Figure 7. Final super light car (SLC)-body multi material concept. Reprinted from Ref. [96].

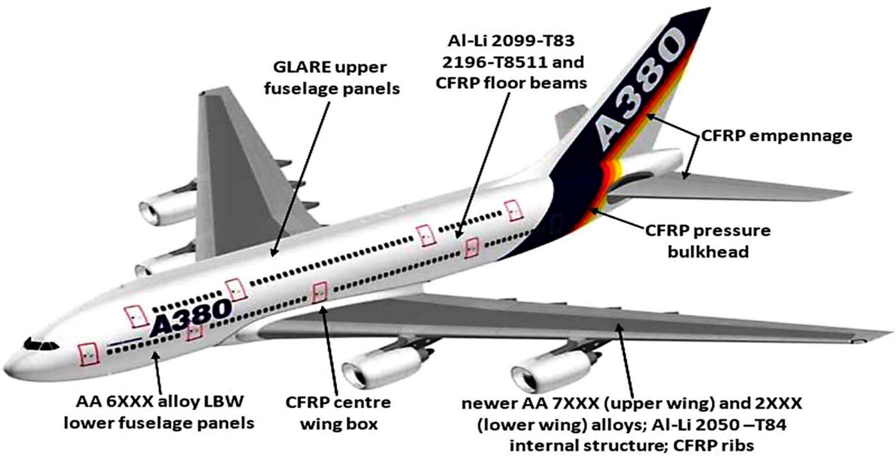

Figure 8. Hybrid metal/composite airframe of aircraft. Reprinted from Ref. [90].

crack propagation under tensile load conditions and results in tensile failure [113]. In one study, Gu et al. [114] worked on the design and cracking susceptibility of additively manufactured Al–Cu–Mg alloys with tandem wires and pulsed arc. It was reported that the heat input during layer deposition has a significant influence on cracking susceptibility. Recently, some studies have shown that high and low cycle fatigues can cause crack initiation, crack growth, and finally ultimate failure within a material [115,116]. However, fatigue has been a dominant mechanical failure mode in aeroengines [117]. High cycle fatigue is regarded as one of the major causes of turbine failures in military aircraft [45]. To avoid mechanical failure in an aeroengine, it is crucial to investigate the effect of fatigue crack initiation (FCI) and short crack growth (SCG) in the aeroengine spare parts produced with the use of additively manufactured aluminium alloys materials [118]. In WAAM techniques, process parameters (such as voltage, wire feed speed, current, inter-pass temperature, and shielding gas flow), material composition, environmental conditions (such as gas contamination) [77], and many other factors can significantly influence oxidation effects. Porosity and cracks have been described as the major dominating setback which often limits the application of aluminium alloys. These defects serve as the location of stress concentration through which the failure of the part begins. The adverse effect of porosity and cracks in WAAM can be completely debased by the optimisation of the process parameters and by employing advanced techniques such as the CMT method approach. According to Sames et al. [104], process parameters can be properly adjusted to avoid a range of mechanisms that can create pores in the Al alloy processing parts. Fu et al. [98] investigated hot wire arc additive manufacturing (HWAAM) of aluminium alloy with reduced porosity and high deposition rate. It was reported that porosity and low deposition rate are two dominating technical challenges in the WAAM of aluminium alloy. It is concluded that the mechanical properties of aluminium components were improved with the reduction of porosity, and the optimal performance of ultimate tensile strength, yield strength, and elongation were 399 MPa, 257 MPa, and 12.0%, respectively.

Duarte et al. [119] proposed a new variant of WAAM based on hot forging. This process involved the use of viscoplastic deformation behaviour at high temperatures to reduce residual stresses, increase ductility, eliminate post-heat treatment operations, and homogenise grain structure. The results demonstrated that the hot forging effect refines the solidification microstructure, thereby improving the yield strength. The yield strength increased from 360 to 450 MPa, ultimate tensile strength improved from 574 to 622, and elongation to fracture reduced from 32 to 28%. The existent internal pores collapsed under the forging force, which is essential for the behaviour in-service of AM parts. Wang et al. [79] examined the effect of pulse frequency and arc current on the gas pores and grains for arc AM Al–5Si alloy. It was reported that coarse grains and gas pores are two major problems that limit the application of AM aluminium alloys. The experiments conducted on pulse frequency and arc current have a significant influence on the macrostructure, microstructure, porosity, and tensile properties of the fabricated samples. The results showed that finely grained microstructure was obtained with low pulse frequency and low arc current as a result of the rapid cooling of the molten pool. Honnige et al. [120] examined the effect of vertical inter-pass rolling and post-deposition side rolling on 2319 aluminium walls fabricated by WAAM techniques. It was demonstrated that vertical inter-pass rolling modifies the residual stress in aluminium WAAM, and it also eliminates the distortion. Moreover, post-deposition side rolling is regarded as an effective method for controlling residual stresses and distortion in additively manufactured aluminium parts with the increase in the hardness by work hardening. Hu et al. [121] worked on a region-based planning method with all horizontal welding positions for robotic curved layer (CL) WAAM. The results suggested that the surface waviness and thickness were relatively consistent at the arbitrary region of the complex surface with the region-based method, and the stair-step effect that appeared in flat layer WAAM was significantly alleviated with the CL-WAAM. Zeng et al. [122] investigated the influence of surface profile on the laser ultrasonic inspection of WAAM defects. The experimental results showed that surface roughness has a significant effect on the signal-to-noise ratio (SNR) of internal through-hole detection, and surface waviness has a significant effect on the sensitivity of superficial crack detection. In one study, Parmar et al. [123] stated that the single-bead geometrical errors produced during the building process of the next layer of material in WAAM can be minimised by incorporating robotic and welding technologies. Xia et al. [124] investigated the anomalies generated during the deposition process, such as humping, spattering, and robot suspending. To recognise these anomalies automatically, convolutional neural networks (CNN) were employed. The results demonstrated that convolutional neural networks (CNN) could obtain high accuracy in classifying melt pool images and be helpful to improve the automation level and production quality for WAAM. Lee et al. [125] also examined the repair of damaged parts of a cross-slide using WAAM in machine tool remanufacturing. It is concluded that the repair of machine tool parts made of grey cast iron was successfully conducted using WAAM with nickel wire. Colegrove et al. [126] investigated the impact of high-pressure rolling on microstructure and residual stress improvement in WAAM parts. It was observed that a slotted roller can limit the amount of lateral deformation providing a more effective reduction in residual stress and distortion when compared with a roller where no lateral restraint is provided. As described in Wang et al. [127], the porosity rate was reduced to 0.1% with the online cooling process, and a decrease in the inter pass temperature during the WAAM process had a significant influence on grain refinement. Köhler et al. [8] studied the effects of the welding process on the build-up accuracy and material properties during WAAM of aluminium structure. In their study, the built samples were analysed in terms of residual stress, surface finishing, and hardness. Residual stress was analysed through X-ray diffraction in two thin-walled specimens each made from 20 layers of either Al-4047 or Al-5356. The inter-pass temperature while welding was 100°C in both samples. It is concluded that the residual stress magnitudes depended on the yield limit of the filler metals used. The first layer of the thin-walled structure as well as the bottom surface of the substrate are in tension, whereas the top layer and substrate surface undergo compression [8]. Figure 9 summarises scanning electron microscope images of the fracture surface morphology in the horizontal direction. The fracture surface was mainly characterised by viewing to studying the trend of a ductile fracture. This implied that the micropores could be detected embedded in the fracture surface morphology [8].

To produce defect-free parts via the WAAM technique, there is a need to address WAAM processing defects during the manufacturing of Al alloy structural parts for the benefit of aerospace and automotive industries. However, many additively manufactured parts in the aerospace and automotive industries are made of aluminium alloys. Hence, the development of WAAM processing techniques that can produce defect-free functional parts in the aerospace and automotive industries is regarded as one of the research directions to improve mechanical properties of WAAM aluminium alloys structures for engineering applications [32].

Summary and future work

WAAM is a promising alternative to conventional subtractive manufacturing for fabricating large high-value metal components. Based on current trends, industry experts have a positive outlook about the future of the WAAM of aluminium alloys in aerospace and automotive industries to enhance automated process planning, monitoring, and control for the WAAM process. The addition of different alloy elements in aluminium in WAAM could reduce the weight of aircraft and improve their structural integrity performance. The combination of conventional subtractive manufacturing and the WAAM process can be further utilised to reduce material wastage and enhance the manufacturing of complex parts for the benefit of aerospace and automotive applications.

This review has discussed the various common defects produced by the application of the WAAM for aluminium alloys which can only be controlled by adopting advanced techniques such as the CMT method, suitable manufacturing process parameters, and post-process treatment to eradicate the caused defects (oxidation, residual stress, cracks, porosity, and distortions) and to improve the quality of the printed parts. In addition, the efforts undertaken by the researchers to mitigate the identified challenges have been documented.

Despite excellent research activities in the aspect of WAAM aluminium alloys, further investigations are necessary to improve and drive the technology-readiness level of WAAM for the production of functional aluminium parts for aerospace and automobile industries. For future work, the following conclusions are drawn from the literature comprehensively reviewed:

(i) Further research is required to investigate the effect of inherent residual stress and distortion generated during WAAM processing as a result of high thermal input from heat sources.

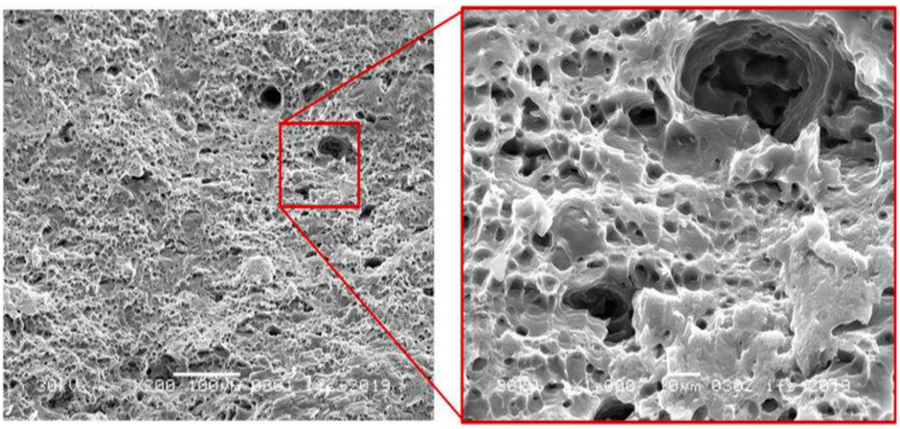

Figure 9. Scanning electron microscope microstructure of the fracture sample after the tensile test. Reprinted from Ref. [8].

(ii) There is a need to determine the suitable power source techniques for a range of materials used in WAAM.
(iii) Further research studies are required to investigate the advanced techniques that can be used to enhance the workability and controllability in GTAW to outperform the conventional GTAW and GMAW processes.
(iv) More investigations into the influence of EAC on the mechanical properties of the corrosion-resistant metallic parts fabricated by using the WAAM method are essential.
(v) More investigations are required to determine the mechanical response of the different combinations of alloying elements in aluminium parts fabricated by WAAM under different loading and operating conditions
(vi) Porosity and cracks have been described as the major setback which often limits the application of aluminium alloys in WAAM. Hence, further investigation is required to determine the appropriate method that can be used for optimising the process conditions to improve the properties of aluminium components deposited by the WAAM process.
(vii) To further enhance product structure, more investigations into the effect of degraded dimensional control and surface integrity of the parts deposited by WAAM are essential.
(viii) To improve the efficiency of the aircraft engine performance, more investigations into the impact of FCI and SCG in the aeroengine spare parts produced with the use of additively manufactured aluminium alloys materials are essential.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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